

ARINDAM SINGHA

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SUMMARY

Robotics and AI researcher with a Ph.D. in Networked Robotics and proven expertise in developing intelligent algorithms, multi-agent control systems for both Unmanned Aerial Vehicle (UAV) and Unmanned Ground Vehicle (UGV) systems, and deep learning solutions. Skilled in computer vision, large language models (LLMs), retrieval-augmented generation (RAG), and advanced model deployment pipelines. Hands-on experience in MLOps, cloud services, and robotics frameworks (ROS2, MoveIt2, NAV2). Recognized for developing scalable AI solutions, integrating ML models into robotic workflows, and driving innovation across research and industry projects. Developed and delivered one microservice-based end-to-end platform to production on vision-based deep learning architecture. Delivered one end-to-end project to production on ML & computer vision.

EDUCATION

Ph.D. (National Institute of Technology Sikkim)

Intelligent Networked Robotic Systems

2016 - 2021

M.E. (Thapar University)

Electronics Instrumentation & Control

2012 - 2014

B.Tech (Abacus Institute of Engineering and Management, MAKAUT)

Applied Electronics and Instrumentation Engineering

2008 - 2012

SKILLS

Reinforcement Learning: MDP, Actor-Critic, PPO, MADT3

Frameworks & Tools: ROS2, Langchain, LangGraph, Triton Inference Service, MCP

ML/DL Expertise: YOLOv3/v4/v8, Transformers, Computer Vision, Semantic Segmentation, Keypoint Detection, LLMs, RAG, AI Agent, Agentic RAG, MCP

Robotics: ROS2, Gazebo, Rviz, URDF, TF2, MoveIt2, NAV2, DDS, LiDAR

Technical Skills: Robotic Control, Computer Vision, Machine Learning, Data Science, Advanced Control Systems

Programming Languages: Python, MATLAB, SCILAB, C

Deployment & MLOps: Docker, Azure Container Registry, Azure Artifacts, GIT, Docker Networking, Azure Cloud, Triton Inference Server, CI/CD pipelines, MLFlow, Dev-Container, DVC, Databricks

Hardware Proficiency: Raspberry Pi, Arduino, Pixhawk Flight Controller

EXPERIENCE

Lead Engineer

Magna Steyr India, Pune

07/2024 – Present

- Developing deep learning-based models to identify defects for multi-class object detection and semantic segmentation using YOLOV8 and encoder-decoder based architecture.

- Developing Python-based data pipelines to efficiently extract and analyze real-time data, and enhancing production monitoring and decision making.
- Deployment of deep learning models using Triton Inference Server with ensemble structure.
- Converted the microservice-based vision platform to a ROS2-based platform and deployed it in production.
- ROS2-based framework development of a three-wheel differential drive vehicle.
- Development of chatbot using agentic-RAG for requirements to test cases development.

Senior Engineer

Magna Steyr India, Pune

02/2022 – 06/2024

- Developed a digital twin of MobileyeQ3 camera sensor in Webots simulation platform.
- Developed ROS2 bridge with Webots to implement the digital twin.
- Developing microservices with gRPC connections for seamless communication.
- Managing deployments on Azure cloud with Docker containerization, ensuring system reliability and scalability.

Senior Research Fellow

National Institute of Technology Sikkim, Sikkim

01/2018 - 12/2021

- Developed an adaptive RBFNN-driven control framework for the formation control of a multi-UAV system under external disturbances.
- Leader-follower-based formation controller design between a multi-UAV system under directed and undirected network topology.

Junior Research Fellow

National Institute of Technology Sikkim, Sikkim

01/2016 - 12/2017

- Developed Neural network-based algorithms for a single Unmanned Ground Vehicle in a grid-based environment.
- Developed a trajectory tracking framework for multiple Unmanned Aerial Vehicles using the Kalman filter.

PROJECTS

Requirement to Test Cases Generation Using LLM

05/2025 - Present

- Built LLM+RAG agent to auto-generate test cases from requirements.
- Created ingestion pipeline and implemented chunking, embedding, and retrieval workflows for CSV, PDF, Excel documents. Along with that added image processing for infotainment UI screenshots.
- Developed chatbot interface for test case refinement and reduced manual test design effort with multimodal inputs.

Traffic Signal Detection

- Developed a computer vision project for traffic signal detection suitable for intelligent transportation or ADAS-oriented experimentation.
- GitHub: https://github.com/arindam-singha/traffic_signal_detection

Realtime PPE Detection

- Built a real-time personal protective equipment detection project for safety monitoring in industrial environments.
- Developed Streamlit UI for real-time model validation.
- GitHub: https://github.com/arindam-singha/Realtime_PPE_Detection

MCP Weather Service

- Built MCP-based weather service for LLM tool-calling and AI agent integration.

- Created Gradio-MCP server UI for tool execution and user interaction.
- GitHub: <https://github.com/arindam-singha/MCP-weather-service>

Development of Vision-Based Platform

05/2023 - 05/2025

- **Previously defect detection taking 10-15 minutes manually, now the entire process is completed within 3secs.**
- Developing deep learning-based models to identify defects for multi-class object detection and semantic segmentation using YOLOV8 and encoder-decoder based architecture.
- Developing Python-based data pipelines to efficiently extract and analyze real-time data, and enhancing production monitoring and decision making.
- Developing microservices with gRPC connections for seamless communication.
- Managing deployments on Azure cloud with Docker containerization, ensuring system reliability and scalability.
- Deployment of deep learning models using Triton Inference Server with ensemble structure.

Benchmarking of Keypoints Detection

07/2024 - 01/2025

- Developing deep learning models using several architectures such as RSN, MSPN, YOLO-NAS to identify keypoints.

Benchmarking Motion Planning Algorithm

07/2024 - 01/2025

- Motion Planning of 6DoF robot manipulator using benchmark ROS2 and MoveIt2 motion planning algorithms.
- Benchmarking different motion planning algorithms such as RRT, RRT Connect for pick and place operation.

Intelligent Networked Robotic Systems

Jan 2016 - Dec 2021

- Developed neural network-based path planning algorithms for multiple Unmanned Ground Vehicles in a grid-based environment.
- Developed an intelligent control framework for multiple Unmanned Aerial Vehicles to achieve and maintain the formation under external disturbances for directed and undirected network topologies.

ACHIEVEMENTS

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- Team of the Month award for the month of October 2023 and April 2025 at Magna Steyr India.
 - Employee of the Month award for the month of December 2022 at Magna Steyr India.
 - Prestigious Visvesvaraya Ph.D. Scholarship under Visvesvaraya Ph.D. Scheme by Digital India Corporation.
 - MHRD GATE Scholarship.

RESEARCH PAPERS

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- Adaptive Neural Network based Quadrotor UAV Formation Control under External Disturbances. *Aerospace Science and Technology* (Springer), Published 2024.
 - Lyapunov based Trajectory Tracking Controller for a Quadrotor UAV with Nonholonomic Constraints. *e-Prime - Advances in Electrical Engineering, Electronics and Energy* (Springer), Published 2024.
 - UAV Formation Control under Fixed and Variable Adjacency Based Directed Network Topologies. *Journal of Scientific and Industrial Research* (NIScPR-CSIR), Published 2022.
 - Leader-follower Based Formation Controller Design for Quadrotor UAVs. *Transactions of the Indian National Academy of Engineering* (Springer), Published 2022.
 - Differentially Flat Trajectory Generation and Controller Design for a Quadrotor UAV. *International Journal of Modelling Identification and Control* (Inderscience), Published 2021.
 - Neural Dynamics Based Complete Grid Coverage by Single and Multiple Mobile Robots. *SN Applied Sciences* (Springer), Published 2021.
 - Path Planning of Mobile Robot using Adaptive Particle Swarm Optimization. *Advances in Intelligent Systems and Computing* (Springer), Published 2021.

- Neural Dynamics based Complete Coverage of Grid Environment by Mobile Robot. *Advances in Intelligent Systems and Computing* (Springer), Published 2021.
- Self-Organizing Map based Strategic Placement and Task Assignment for a Multi-agent System. *Advances in Intelligent Systems and Computing* (Springer), Published 2021.
- UGV-UAV Coordination: Take-off, Landing and Formation Control. *World Automation Congress (WAC)* (IEEE), Published 2021.
- Trajectory Tracking in the Desired Formation around a Target by Multiple UAV Systems. *Procedia Computer Science* (Elsevier), Published 2018.
- Grid-based UGV Navigation in a Dynamic Environment using Neural Network. *International Conference on Inventive Research in Computing Applications (ICIRCA)* (IEEE), Published 2018.
- Navigation of Mobile Robot in a Grid-based Environment using Local and Target Weighted Neural Networks. *8th International Conference on Computing, Communication and Technologies (ICCCNT)* (IEEE), Published 2017.